Final report on the emotion classifier project

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1. Introduction

How do humans communicate?

Most people would immediately think of spoken or written language. Yet communication goes beyond words. An essential component of human interaction is *emotion*. Emotions form a key part of nonverbal communication and have been widely studied within the field of psychology.

1.1. What is facial emotion recognition

Facial emotion recognition (FER) refers to the ability to identify human emotions computationally based on images. Other types of emotion recognition tasks approach the same problem by analyzing speech or writing to reconstruct a person's emotional state. As past research has shown, understanding human emotions is no trivial task for an algorithm. The best results in the field of FER have been achieved using complex model architectures and significant amounts of training data. However, the results are quite promising. Recent publications have exceeded accuracies of 90% [Citation] in their studies.

1.2. Why is FER important

Given that humans themselves are capable of emotion recognition, it may not be immediately clear why research in this field is so important. The answer to this question is that characterizing a person's feelings is a key part of many important sectors such as:

- Healthcare
- Public safety
- Crime detection
- Advertising

[Citation]

By automating the emotion recognition process needed in these areas, the risk of human error can be significantly reduced. Furthermore individuals differ substantially in their ability to correctly classify sentiments. Implementing automated emotion recognition solutions therefore provides a consistent baseline for real-world applications.

1.3. Our work

Our group was tasked to research and understand this technology, followed by the construction, training and evaluation of our own AI model specialized in assessing six basic human emotions: happiness, anger, sadness, fear, disgust and surprise. This paper provides an insight into the most important parts of the development process (Datasets, Data preprocessing, Model architecture), the achieved results and further experiments.

References